

Thermodynamics Formula Sheet

Ch 4: Energy Analysis of Closed Systems

Units and Conversions

SI Prefixes

femto (f)	$= 10^{-15}$	deca (da)	$= 10^1$
pico (p)	$= 10^{-12}$	hecto (h)	$= 10^2$
nano (n)	$= 10^{-9}$	kilo (k)	$= 10^3$
micro (μ)	$= 10^{-6}$	mega (M)	$= 10^6$
milli (m)	$= 10^{-3}$	giga (G)	$= 10^9$
centi (c)	$= 10^{-2}$	tera (T)	$= 10^{12}$
deci (d)	$= 10^{-1}$	peta (P)	$= 10^{15}$

Atmospheric Pressure

$$\begin{aligned}
 1 \text{ atm} &= 101.325 \text{ kPa} \\
 &= 1.01325 \text{ bar} \\
 &= 760 \text{ Torr} \\
 &= 760 \text{ mmHg} \\
 &= 29.92 \text{ inHg} \\
 &= 14.696 \text{ psi} \\
 &= 10.33 \text{ m H}_2\text{O (at } 4^\circ\text{C)}
 \end{aligned}$$

Pressure Conversions

$$\begin{aligned}
 1 \text{ Pa} &= 1 \text{ N/m}^2 \\
 1 \text{ bar} &= 10^5 \text{ Pa} \\
 1 \text{ psi} &= 6.89476 \times 10^3 \text{ Pa} \\
 1 \text{ Torr} &= 133.322 \text{ Pa} \\
 1 \text{ mmHg} &= 133.322 \text{ Pa} \\
 1 \text{ inHg} &= 3.38639 \times 10^3 \text{ Pa}
 \end{aligned}$$

Gauge vs Absolute Pressure

$$P_{\text{gauge}} = P_{\text{absolute}} - P_{\text{atmospheric}}$$

$$P_{\text{vacuum}} = P_{\text{atmospheric}} - P_{\text{absolute}}$$

Note: Generally, gauge pressures already account for atmospheric pressure, and thus reads zero when open to the atmosphere.

Temperature Conversions

- Absolute temperature conversions
 - $T_K = T_{\circ C} + 273.15$
 - $T_{\circ R} = T_{\circ F} + 459.67$
 - $T_{\circ F} = \frac{9}{5}T_{\circ C} + 32$
 - $T_{\circ R} = \frac{9}{5}T_K$
- Temperature difference conversions
 - $\Delta K = \Delta ^\circ C$
 - $\Delta ^\circ R = \Delta ^\circ F$
 - $\Delta ^\circ F = \frac{9}{5}\Delta ^\circ C$
 - $\Delta ^\circ R = \frac{9}{5}\Delta K$

Density and Specific Gravity

$$\rho = \frac{m}{V} \quad (\text{density}), \quad \vartheta = \frac{1}{\rho} = \frac{V}{m} \quad (\text{specific volume})$$

Specific gravity is defined as a relative density compared to water (at 4°C) where $\rho_{\text{water}} = 1000 \text{ kg/m}^3$:

$$SG = \frac{\rho_{\text{fluid}}}{\rho_{\text{water}}} \implies \rho_{\text{fluid}} = SG \cdot \rho_{\text{water}}$$

Specific Weight

$$\gamma_s = \rho g \quad (\text{specific weight})$$

Where hydrostatic pressure can be written as $P = \gamma_s h$

Volume

$$1 \text{ m}^3 = \underline{1000 \text{ L}} = 61024 \text{ in}^3 = 35.315 \text{ ft}^3 = \underline{264.17 \text{ gal}}$$

Slugs and lbm

$$\left[\frac{1 \text{ lbf}}{32.174 \frac{\text{lbm} \cdot \text{ft}}{\text{s}^2}} \right], \quad \left[\frac{1 \text{ slug}}{32.174 \text{ lb}} \right], \quad \left[\frac{1 \text{ lbf}}{1 \frac{\text{slug} \cdot \text{ft}}{\text{s}^2}} \right]$$

Energy Units

$$1 \text{ Btu} = 778.169 \text{ ft} \cdot \text{lbf} = 1055.06 \text{ J}$$

Work Units

$$1 \text{ hp} = 550 \text{ ft} \cdot \text{lbf/s} = 745.7 \text{ W}$$

Distance

$$1 \text{ mile} = 5280 \text{ ft}$$

Ideal Gas Equation of State

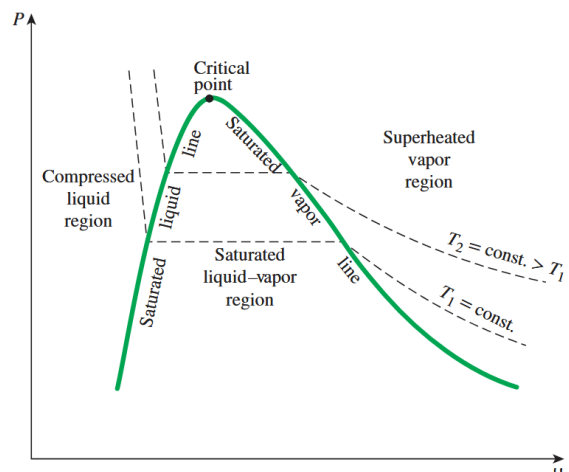
Refer to table A-4 for gas constants of common gases.

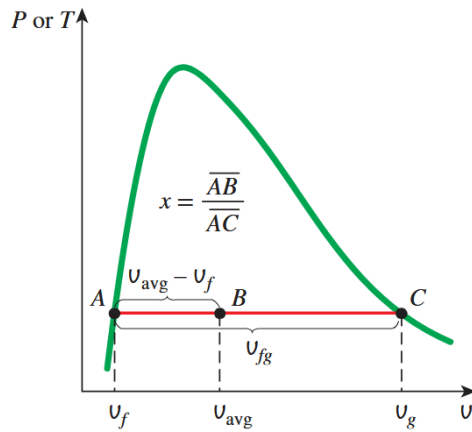
$$P\vartheta = RT, \quad R = \frac{\bar{R}}{M} \quad \left(\text{kJ/kg} \cdot \text{K} \quad \text{or} \quad \frac{\text{Btu}}{\text{lbm} \cdot \text{R}} \quad \text{or} \quad \frac{\text{psia} \cdot \text{ft}^3}{\text{lbm} \cdot \text{R}} \right)$$

$$P\bar{V} = mRT \quad (\text{in terms of TOTAL volume and mass})$$

Universal gas constant: $\bar{R} = 8.314 \text{ J}/(\text{mol} \cdot \text{K})$

Note: Convert units for P , $\bar{V}$, ϑ , and T to match the units of R when using the ideal gas law. Also, $[\text{kJ/kPa}] = [\text{m}^3]$





Phase Change & Saturation

$$h_{fg} = h_g - h_f \quad (\text{latent heat of vaporization})$$

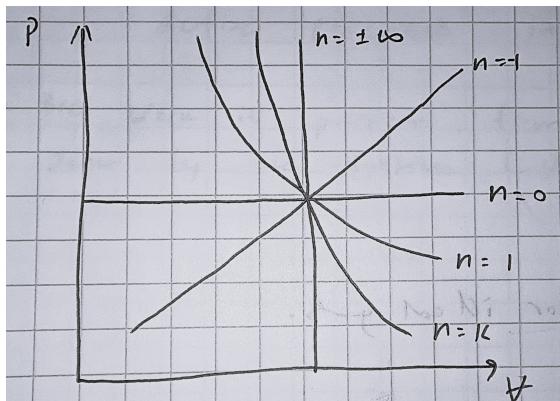
$$x = \frac{m_g}{m_f + m_g} \quad (\text{quality definition})$$

$$\left. \begin{aligned} y_{\text{avg}} &= y_f + x(y_g - y_f) \quad (\text{for } \varphi, h, u) \\ x &= \frac{y_{\text{avg}} - y_f}{y_g - y_f} \quad (\text{solve for quality}) \end{aligned} \right\} y_{fg} = y_g - y_f$$

Compressed Liquid Approximation

$$y(T, P) \approx y_f(T) \quad (\text{use saturated liquid at same } T)$$

Polytropic Processes



$$Pv^n = \text{constant}$$

- $n = 0$: isobaric process (constant pressure)
- $n = 1$: isothermal process (constant temperature, ideal gas)
- $n = k$: isentropic process (constant entropy)
- $n = \infty$: isochoric process (constant volume)

Pressure Relations

$$P = \frac{F}{A} = \rho gh \quad (\text{hydrostatic pressure})$$

$$P_1 = P_2 \Rightarrow \frac{F_1}{A_1} = \frac{F_2}{A_2} \quad (\text{Pascal's Law})$$

$$\Delta P = \rho gh \quad (\text{multi-fluid manometer})$$

(add downward, subtract upward)

Energy Definitions

$$e = u + \frac{1}{2}v^2 + gh \quad (\text{specific total energy})$$

$$e_{\text{mech}} = \frac{P}{\rho} + \frac{1}{2}v^2 + gh \quad (\text{specific mechanical energy})$$

First Law (Closed System)

$$\Delta U = Q - W \quad (\text{energy balance for closed system})$$

$$\dot{E}_{\text{in}} - \dot{E}_{\text{out}} = \frac{dE_{\text{system}}}{dt} \quad (\text{rate form})$$

First Law (Open System)

$$\frac{dE}{dt} = \dot{Q} - \dot{W} + \underbrace{\sum \dot{m}_i \left[h_i + \frac{1}{2}V_i^2 + gz_i \right]}_{\text{Inlet}} - \underbrace{\sum \dot{m}_e \left[h_e + \frac{1}{2}V_e^2 + gz_e \right]}_{\text{Outlet}}$$

For steady state process with no heat transfer:

$$\dot{W} = \dot{m}e_{\text{mech}}$$

$$\dot{W} = \dot{m} \frac{\Delta P}{\rho} = \dot{V} \Delta P \quad (\text{pump power w/ no change in KE or PE})$$

Mass & Flow Relations

$$\dot{m} = \rho \dot{V} = \rho A v_{\text{avg}} \quad (\text{mass flow rate})$$

$$\dot{E} = \dot{m}e \quad (\text{energy flow rate})$$

Enthalpy

$$h = u + P\varphi, \quad \varphi = \frac{1}{\rho} \quad (\text{specific enthalpy})$$

Work

$$W = \int F \cdot dx, \quad \dot{W} = F \cdot v \quad (\text{general})$$

$$\begin{aligned} W &= \int P dV \Rightarrow W = \frac{P_2 V_2 - P_1 V_1}{-n + 1} \quad (\text{polytropic, } n \neq 1) \\ &\Rightarrow W = C \ln \left(\frac{V_2}{V_1} \right) \quad (\text{isothermal, } n = 1) \\ &\Rightarrow W = P \Delta V \quad (\text{isobaric}) \end{aligned}$$

Where C is $P_1 V_1^n = P_2 V_2^n = C$ for a polytropic process.

$$W = \frac{1}{2}k(x_2^2 - x_1^2) \quad (\text{spring work})$$

Heat Transfer

$$Q = \int \dot{Q} dt = \dot{Q} \Delta t \quad (\text{total heat transferred})$$

$$\dot{Q} = kA \frac{\Delta T}{\Delta x} \quad (\text{conduction}), \quad \dot{Q} = hA(T_s - T_\infty) \quad (\text{convection})$$

$$\dot{Q} = \epsilon \sigma A (T_s^4 - T_\infty^4) \quad (\text{radiation})$$

Specific Heat Capacity

$$c_\varphi = \left(\frac{\partial u}{\partial T} \right)_\varphi \quad (\text{fixed volume}), \quad c_p = \left(\frac{\partial h}{\partial T} \right)_p \quad (\text{fixed pressure})$$

$$\Rightarrow u_2 - u_1 = c_\varphi [T_2 - T_1] \quad (\text{constant } c_\varphi \text{ or } c_p)$$

$$c_p = c_\varphi + R \quad (\text{ideal gas}), \quad c_p = c_\varphi = c \quad (\text{incompressible substances})$$

$$Q = mc\Delta T \quad c_{\text{water}} = 4.18 \frac{\text{kJ}}{\text{kg} \cdot \text{K}}$$

$$k = c_p / c_\varphi \quad (\text{specific heat ratio})$$